

ERYTHROBLASTOSIS FETALIS

(Hemolytic Disease of the Fetus and Newborn - HDFN)

- **Definition :** Hemolytic disease of fetus and newborn characterized by abnormal hemolysis of RBCs.
- **Cause :** Due to **Rh incompatibility** (*difference between Rh blood group of mother and baby*).
- Characterized by the presence of **erythroblasts in peripheral blood**.

PATHOGENESIS OF Rh INCOMPATIBILITY

1st Pregnancy (Mother Rh-ve, Fetus Rh+ve)

1st Child escapes complications
(Rh antigen cannot cross placental barrier during pregnancy)

During Parturition:
Placental detachment causes fetal Rh+ RBCs to leak into maternal blood

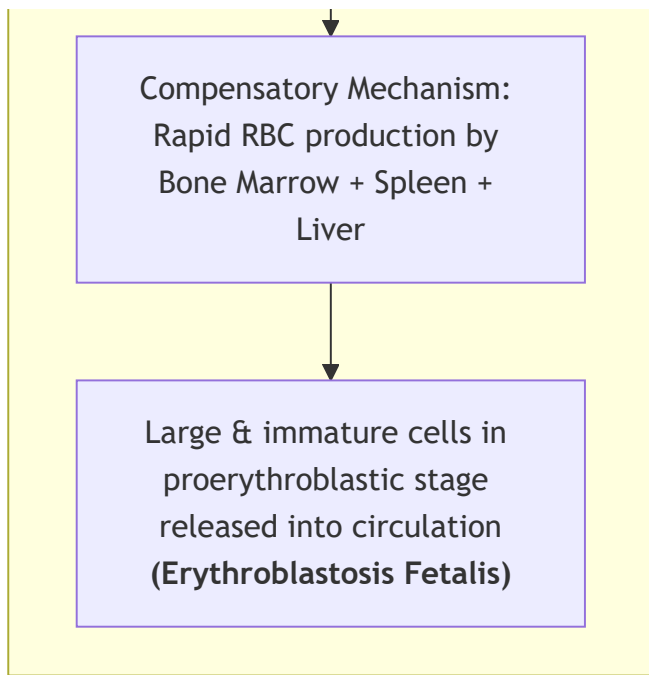
Within 1 month after delivery: Mother develops anti-Rh antibodies (Anti-D)

2nd Pregnancy (Fetus Rh+ve again)

Maternal anti-Rh antibodies (IgG) cross placental barrier into fetal circulation

Agglutination and hemolysis of fetal RBCs

Jaundice & severe anemia



SEVERE COMPLICATIONS

- Due to excessive hemolysis, severe complications develop:
 - a. **Severe Anemia**
 - b. **Hydrops Fetalis**
 - c. **Kernicterus**

1. Severe Anemia

- Excessive hemolysis → Severe anemia → **Infant dies.**

2. Hydrops Fetalis

- A serious condition in fetus characterized by **generalized edema**.
- Severe condition leading to **intrauterine death of fetus**.

3. Kernicterus

- Form of brain damage in infants caused by **severe jaundice** (*high unconjugated bilirubin content*).

- **Blood-Brain Barrier (BBB)** is not well developed in infants compared to adults.



- Bilirubin enters brain and causes **permanent brain damage**.

Commonly Affected Brain Regions :—

- a. Basal ganglia
- b. Hippocampus
- c. Geniculate bodies
- d. Cerebellum
- e. Cranial nerve nuclei

PREVENTION & TREATMENT

1. Prophylaxis (Anti-D Administration)

- **During Gestation** : Anti-D immunoglobulin administered to Rh-negative mother at **28th – 34th weeks of gestation**.
- **Postpartum** : If mother delivers an Rh +ve baby, **anti-D is administered within 48 hours**:

Anti-D administered



Develops passive immunity



Destroys fetal Rh+ RBCs before mother's immune system reacts



Prevents active formation of Rh antibodies in mother's blood

2. Treatment of Affected Infant (Exchange Transfusion)

- If baby is born with erythroblastosis fetalis:

Treatment given by means of **Exchange Transfusion**



Rh -ve blood transfused into infant (replacing infant's own Rh +ve blood)



Takes at least **6 months** for infant's new Rh +ve blood to replace transfused Rh -ve blood



By this time, all maternal anti-Rh antibodies in infant's body are completely destroyed